Author: Strictly speaking, it does. A cosine defined, for example, on the real line is yet another function. In other words, using cosine we may, if we wish, define any number of different functions by varying the domain of these functions.

In the most frequent case, when the domain of a function coincides with the domain of an analytical expression for the function, we speak about a natural domain of the function. Note that in the examples in the previous dialogue we dealt with the natural domains of the functions. A natural domain is always meant if the domain of a function in question is not specified (strictly speaking, the domain of a function should be specified in every case).

Reader: It turns out that one and the same function can be described by different formulas and, vice versa, one and the same formula can be used to "construct" different functions. → V. Imp

Author: In the history of mathematics the realization of this fact marked the final break between the concept of a function and that of its analytical expression. This actually happened early in the 19th century when Fourier, the French mathematician, very convincingly showed that it is quite irrelevant whether one or many analytical expressions are used to describe a function. Thereby an end was put to the very long discussion among mathematicians about identifying a function with its analytical expression.

It should be noted that similarly to other basic mathematical concepts, the concept of a function went through a long history of evolution. The term "function" was introduced by the German mathematician Leibnitz late in the 17th century. At that time this term had a rather narrow meaning and expressed a relationship between geometrical objects. The definition of a functional relationship, freed from geometrical objects, was first formulated early in the 18th century by Bernoulli. The evolution of the concept of a function can be conventionally broken up into three main stages.

- During the first stage (the 18th century) a function was practically identified with its analytical expression.
- During the second stage (the 19th century) the modern concept of a function started to develop as a mapping of one numerical set onto another.